

DATA STRUCTURE

LAB PROGRAM

Write a C program to arrange a series of numbers using Merge Sort

```
#include <stdio.h>
```

```
void merge(int a[], int low, int mid, int high)
```

```
{
```

```
    int temp[100];
```

```
    int i = low, j = mid + 1, k = low;
```

```
    while(i <= mid && j <= high)
```

```
    {
```

```
        if(a[i] < a[j])
```

```
            temp[k++] = a[i++];
```

```
        else
```

```
            temp[k++] = a[j++];
```

```
    }
```

```
    while(i <= mid)
```

```
        temp[k++] = a[i++];
```

```
    while(j <= high)
```

```
temp[k++] = a[j++];
```

```
for(i = low; i <= high; i++)
```

```
    a[i] = temp[i];
```

```
}
```

```
void mergesort(int a[], int low, int high)
```

```
{
```

```
    int mid;
```

```
    if(low < high)
```

```
    {
```

```
        mid = (low + high) / 2;
```

```
        mergesort(a, low, mid);
```

```
        mergesort(a, mid + 1, high);
```

```
        merge(a, low, mid, high);
```

```
    }
```

```
}
```

```
int main()
```

```
{
```

```
int a[100], n, i;
```

```
printf("Enter number of elements: ");
```

```
scanf("%d", &n);
```

```
printf("Enter elements:\n");
```

```
for(i = 0; i < n; i++)
```

```
    scanf("%d", &a[i]);
```

```
mergesort(a, 0, n - 1);
```

```
printf("Sorted array:\n");
```

```
for(i = 0; i < n; i++)
```

```
    printf("%d ", a[i]);
```

```
return 0;
```

```
}
```

Output

Enter number of elements: 5

Enter elements:

5 55 5555555 5555 55555

Sorted array:

5 55 5555 55555 5555555

=== Code Execution Successful ===